CS Capstone Design

Product Demonstration Grading Sheet (100 pts)

TEAM: TJ Davis, Clayton Ramsey, Lucas Larson, Victor Rodriguez

PROJECT: REACH

DEVELOPMENT CYCLE: ☐ Alpha Demo I ☒ Alpha Demo II ☐Acceptance Demo

DATE: 1/20/2026

MENTOR: Bailey Hall

CLIENT: Dr. Lerner and Dr. da Cunha

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1. Overview

The purpose of the Product Demonstration Flight Plan is to define what functionality is planned, what functionality will be demonstrated in the current development cycle, and how success will be evaluated.

This document serves three roles:

1. A **planning artifact**, showing how the full product scope is distributed across development cycles
2. A **demonstration guide**, outlining how the current cycle's functionality will be presented through realistic user flows
3. An **evaluation aid**, defining acceptance criteria used by the mentor and client to assess completeness and correctness

The Flight Plan is prepared by the team, reviewed and approved by the mentor, and used during the live demonstration for evaluation and feedback.

2. Product Scope and Development Cycles Overview

This section presents the full set of user stories or use cases for the product and how they are distributed across the three development cycles. The intent is to make clear:

- What functionality exists in the overall product, corresponding to your full prioritized list of functional requirements
- What is included in each development cycle
- How incremental delivery builds toward the final system

2.1. Backlog Overview by Development Cycle

Alpha Demo I – Current Cycle

ID	User Story / Use Case	Acceptance Criteria (Summary)
US-1	As a user, I want to launch a simulated environment so that I can observe a system operating in a virtual setting.	MuJoCo scene loads without errors; the robotic arm appears correctly positioned relative to the user model; simulation runs and renders consistently.
US-2		The arm responds to control inputs; joints move within defined limits; no instability or simulation crashes occur.

	As a user, I want the robot to execute basic joint movements so that I can verify control and kinematics.	
US-3	As a researcher, I want to train a reinforcement learning agent on a simple reaching task so that I can confirm the learning pipeline works end-to-end.	Training script runs successfully; reward values update during training; trained policy produces non-random arm motion toward a target.
US-4	As a developer, I want to visualize training and evaluation rollouts so that I can validate agent behavior qualitatively.	Evaluation script renders rollouts; end-effector trajectory is visible; logs confirm successful episode execution.

Alpha Demo II - Planned Scope

ID	User Story / Use Case	Acceptance Criteria (Summary)
US-1	As a user, I want the robotic arm to reliably reach a defined target with full body motion enabled in all the joints.	The end-effector reaches within a defined distance threshold of the target in a majority of evaluation episodes.
US-2	As a researcher I want the framework to be easily adaptable to multiple robot types and scenes.	The framework has relatively easily swappable robots, and a new scene or robot can be uploaded and tested by a researcher with minimal technical skills needed.
US-3	As a researcher, I want the AINEX robot to be able to recognize a user gesture and report back to the user.	The ainex robot is able to use computer vision to determine if a human hand is in front of its camera and correctly identify a “high five” gesture.

Acceptance Demo - Planned Scope

ID	User Story / Use Case	Acceptance Criteria (Summary)
US-1	As an end user, I want the robotic arm to perform a successful reaching interaction in a wearable context so that its assistive potential is demonstrated.	The agent consistently reaches the target in evaluation mode; motion appears safe and controlled; no collisions or instability occur.

US-2	As a researcher, I want the trained policy to generalize to new target positions so that it demonstrates robustness.	Policy succeeds across multiple target locations not seen during training; success rate meets predefined threshold.
US-3	As a reviewer, I want a live demonstration of the system running end-to-end so that I can assess project completeness.	Demo runs without manual intervention; simulation loads, policy executes, and results are visible within the demo timeframe.
US-4	As a project stakeholder, I want documentation of system design and limitations so that readiness for sim-to-real transfer can be evaluated.	Architecture, assumptions, and known limitations are clearly presented; risks and next steps toward physical deployment are explained.

3. Demonstration Walkthrough

For each user story or use case included in the current cycle, provide a demonstration walkthrough guided by acceptance criteria. The demo should follow a realistic user flow and show the functionality working end-to-end.

Use Case / User Story: US-1 Reaching Task

User Role: End User

Goal: Demonstrate reaching task with all joints enabled

Demonstration flow:

1. **Open MuJoCo simulation window using the render demo scripts**
2. **Demonstrate the simulation is live and can model**
3. **Show that the model can reach the target with all joints enabled**
4. **Close the simulation**

Acceptance Criteria (What Success Looks Like)

The demonstration is considered successful if:

1. **It can run the simulation robot and have the correct viewer window**
2. **The end effector of the robot is able to reach the target at least once with all joints working**

Evaluation and Comments (for mentors only):

- Convincingly demonstrated the functionality

- Other evaluative comments/notes

Use Case / User Story: US-2 Framework adaptability

User Role: Researcher

Goal: The framework is able to support uploads of different robots

Demonstration flow:

1. A new robot is downloaded from the internet

2. Files are uploaded to the project

3. Using YAML config, the robot path is specified

4. The render scene script is ran

5. Using YAML config, a a different scene is specified

6. The render scene script is ran again and renders the different scene

Acceptance Criteria (What Success Looks Like)

The demonstration is considered successful if:

1. **Simulation opens for a specified scene from the YAML config**
2. **A different robot is rendered from the specified config**
3. **A different scene is rendered from the specified config files**

Evaluation and Comments (for mentors only):

- Convincingly demonstrated the functionality

- Other evaluative comments/notes

Use Case / User Story: US-3 Gesture Recognition

User Role: Researcher

Goal: Recognize a "High Five" Gesture

Demonstration flow:

1. **Begin camera module script on AINEX**
2. **Demonstrate that there is no output when there is no gesture present**
3. **Present a High Five gesture in front of AINEX camera**
4. **Record an output for a positive response to the gesture**

Acceptance Criteria (What Success Looks Like)

The demonstration is considered successful if:

1. **Gesture data is fed to the robot**

2. **Textual confirmation is returned from the recognition model**

Evaluation and Comments (for mentors only):

- Convincingly demonstrated the functionality

- Other evaluative comments/notes

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4. Out-of-Scope or Deferred Functionality

List any user stories or use cases planned in the approved flight plan for this cycle, but not demonstrated, along with their planned future cycle. This helps set clear expectations and avoids confusion during the demo.

Alpha Demo 2 – Current Cycle

ID	Reason Not Demonstrated	Planned Cycle

5. Known Limitations and Planned Follow-Up

Document any known limitations (if any), partial implementations, or issues observed during development or demonstration. Where possible, indicate planned mitigation or timing for resolution.

- Limitation / Issue: New robotic model is unstable
- Impact: Low impact
- Planned Resolution (if known): We will reuse our old model for user stories that require it.

6. Approval

This Flight Plan has been reviewed and approved for use in the Alpha Demonstration.

Mentor Approval: Bailey Hall

Date: 03/16/2026